

# Nathan Potraz

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## Education

**Bachelor of Science in Computer Science** | California State, San Marcos • 3.7 GPA • Cum Laude

**Relevant Coursework:** OOP, Data Structures & Algorithms, Networking, Cloud, Web Programming, Databases, Computer Security, AI, Operating Systems, AGILE/SCRUM.

## Projects

**Devlog** | [devlog.ntpotraz.dev](https://devlog.ntpotraz.dev)

Oct 2025 – Present

- A journaling web app built using **TypeScript React** and a **Golang HTTP server and API**.
- Features simple and fast journal entry **creation, sharing, and editing** utilizing **SQLite hosting**.
- Integrated Clerk for **secure user authorization**, supporting GitHub and Google SSO
- Automated deployment with **GitHub Actions** and **Docker**, reducing production release time **from 15 minutes to under 3 minutes** via a **CI/CD pipeline** targeting Google Cloud Run.

**PIC Health & Resource Mobile App** | [ntpotraz.github.io/PIC-Health-App](https://ntpotraz.github.io/PIC-Health-App)

Aug 2024 – Dec 2024

- **Led a 5 developer team** to architect and implement the core UI/UX for a cross-platform community resource hub using **React Native**.
- Served as **Technical Liaison between developers and stakeholders**, translating requirements into actionable sprint goals that increased development velocity by 20%.
- Engineered a **serverless architecture utilizing the Google Calendar API** and Forms, reducing infrastructure costs to \$0 and enabling non-technical clients to manage events seamlessly.

**Japanese-English Translator** | [github.com/ntpotraz/cs421-translator](https://github.com/ntpotraz/cs421-translator)

Mar 2024 – May 2024

- **Engineered a C++ natural language processor** using a **Deterministic Finite Automata Scanner** to tokenize and categorize complex linguistic inputs with 100% lexical accuracy.
- Built a **Recursive Descent Parser** to enforce strict grammar rules, implementing a **look-ahead mechanism** to handle multi-token syntax structures efficiently.
- Utilized **dictionary lookups for an O(1) average time complexity** by implementing **Hash Maps**, enabling near-instant translation of large text files compared to standard linear search methods.
- Developed a **translation engine** that programmatically reordered tokens to account for syntactic differences between Japanese and English, ensuring grammatically coherent output.

**Gloomy Monsters** | [github.com/ntpotraz/gloomy-monsters](https://github.com/ntpotraz/gloomy-monsters)

Mar 2020 – Mar 2020

- Created a **React-based app** to **manage complex board game mechanics** and stat tracking for users.
- Designed a **nested state architecture** to handle **hierarchical data** for multiple active monster groups.
- Integrated a **complex JSON dataset** to **dynamically populate unit statistics** and deck behaviors.
- Applied **conditional visual styling** to adjust UI elements based on specific game rules and unit types.

## Digital Input Controller Design

**Hardware & Software Engineer** | Self-Employed, Oceanside, CA

June 2022 – Present

- Engineered **over 100 custom-tuned** competitive gaming controllers, performing precise circuit-board soldering and firmware optimization to meet professional-grade performance standards.
- **Consulted directly with clients** to define technical requirements, translating user performance needs into custom hardware and software specifications.
- **Managed full-cycle product development**, including supply chain logistics, hardware debugging, and providing post-deployment technical support
- **Contributed to an open-source motherboard project** by performing board testing, taking electrical measurements, and authoring technical documentation to improve community hardware standards

## Skills

**Backend:** C++, Python, Golang, Java, SQL, SQLite, MySQL, PostgreSQL

**Frontend:** TypeScript, JavaScript, PHP, React, React Native

**DevOps/Tools:** Linux, Docker, CI/CD, Git, GitHub, Agile, AWS, GCP